



Statistics

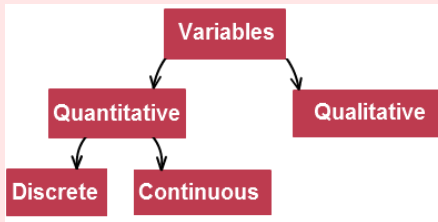
Lectures

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Variable



Variable is a characteristic that vary from one element to another element. For example, Income, score, taste, temperature etc. There are two types of variables, qualitative and quantitative variables.

Types of Variable

Quantitative variable

Quantitative variables are those which represent quantity of something. These variables assign numeric values in them. Quantitative variables are the variables that describe: how much or how many. For example: Marks obtained, height and weight. These variables are quantitative since they assign only numerical data.

Discrete variable

A variable whose values are countable is called discrete variable. Discrete variable assume integer value over a certain interval. For example, marks of students, number of sales man in a company etc.

Continuous variable

A variable that can assume any numerical value over a certain interval. For example, height and weight are of students of management department are continuous variables

Qualitative Variables

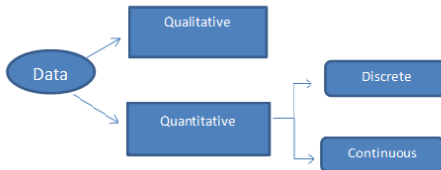
A variable that cannot assume numerical values, but can be classified into two or more non-numeric categories is called qualitative or categorical variable. For example color, gender of a person etc

Example

Data from a pharmaceutical company medical study contain values of many variables for each of the people who were the subjects of the study. Which of the following variables are categorical and which are quantitative?

- (a) Gender(female or male)
- (b) Age(years)
- (c) Race(Asian,black,white,or other)
- (d) Smoker(yes or no)
- (e) Systolic blood pressure(millimeters of mercury)
- (f) Level of calcium in the blood(micrograms per milliliter)

Data and its types



A data set is a collection of information on one or more variable. For example, marks of 40 students in mathematics, opinion of 100 voters.

Quantitative and Qualitative data

Quantitative data are observations measured on a numerical scale. Non numerical data that can only be classified into one of the groups of categories are said to be qualitative or categorical data

Example

Data on response to a particular therapy could be classified as no improvement, partial improvement, or complete improvement. These are qualitative data. The number of minority-owned businesses in Florida is quantitative data. The marital status of each person in a statistics class as married or not married is qualitative or categorical data. The number of car accidents in different U.S. cities is quantitative data. The blood group of each person in a community as O, A, B, AB is qualitative data

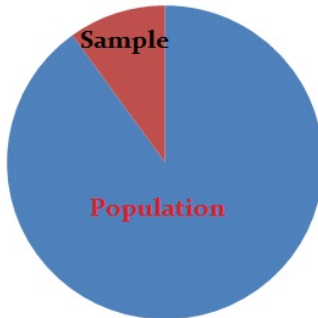
Primary data and its Sources

Data observed or collected directly from first-hand experience.

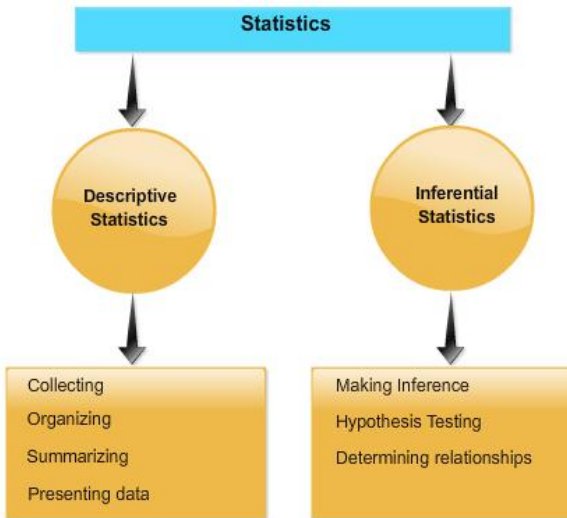
Secondary data and its Sources

Published data and the data collected in the past or other parties is called secondary data

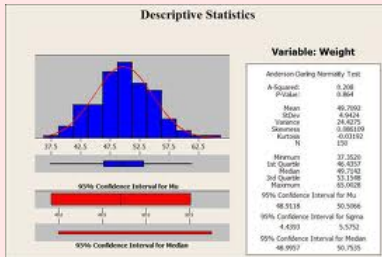
Population and Sample



Statistics

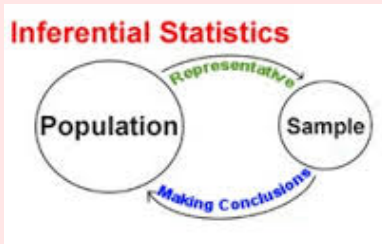


Descriptive Statistics



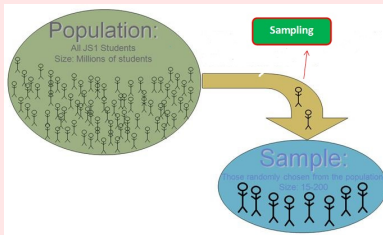
In descriptive statistics, we deal with collection of data, its presentation in various forms, such as tables, graphs and diagrams and findings averages and other measures which would describe the data.

Inferential Statistics



In inferential statistics, we study the techniques used for analysis of data, making the estimates and drawing conclusions from limited information taken on sample basis and testing the reliability of the estimates

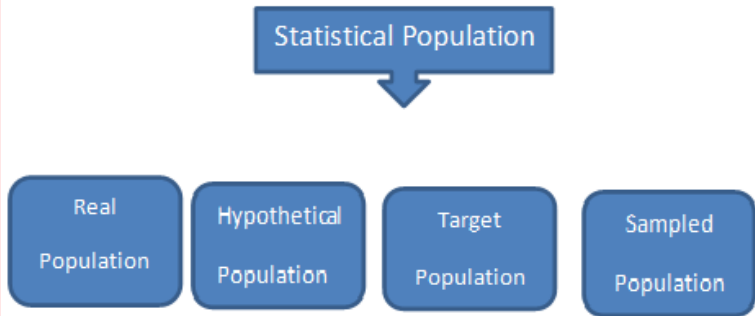
Population, Sample and Sampling



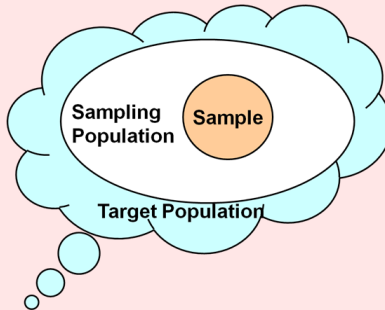
A population is a collection of objects or units about which we want to know something or draw an inference

A sub set of the population, which represents the entire population, is called a sample. Sample is a representative part of the population. The sample is denoted by s and its size by n

Types of Population



Target and Sampled Population



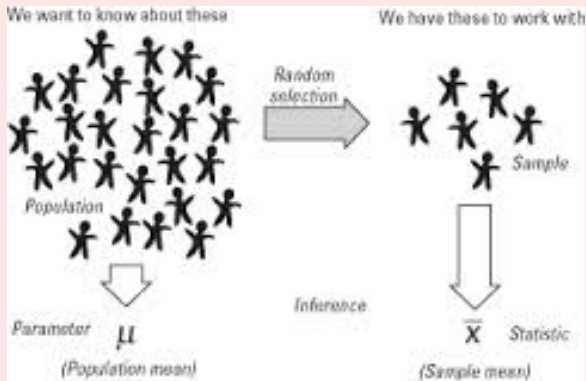
Example

Roger Amster teaches an English course in which 40 students are enrolled. After yesterday's class, Roger questioned the 5 students who always sit in the back of the classroom. Three of the 5 said "yes" when asked if they would like *A Tale of Two Cities* as the next class reading assignment.

- Identify the population and the sample in this situation.
- Is this likely to be a representative sample? If not, why not?

Introduction

Parameter and statistic



A numerical value that describes the characteristic of a population is known as parameter. Parameter is a fixed quantity. A numerical value that describes the characteristic of a sample is known as Statistic. Statistic is not fixed but vary from sample to sample

Parameter and Statistic

Variable



1 2 3 4 5

Statistic



Average = 3.75

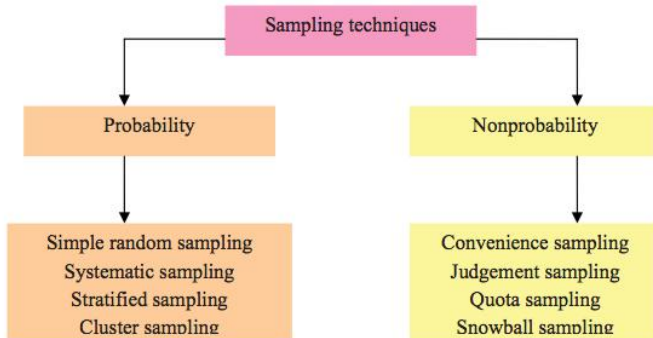
Parameter



Average = 3.72

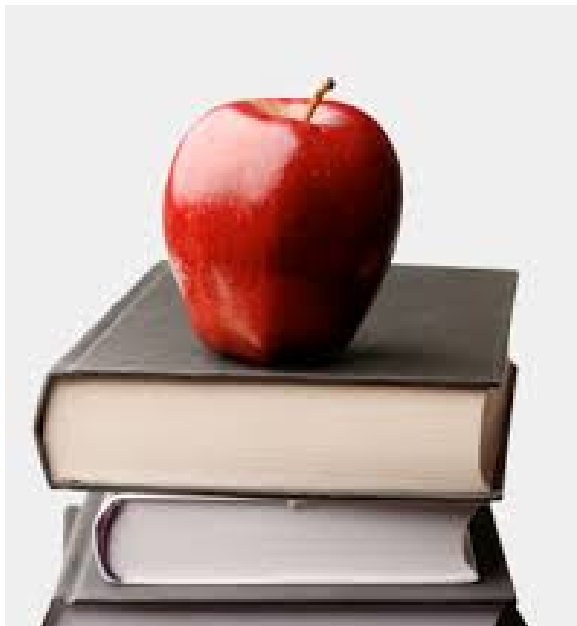
Sampling

Classification of sampling techniques



The process of selecting a sample from the population is known as sampling. There are two types sampling, probability / random sampling and non-probability / non random sampling.





Presentation of Data

The following data show the marks of students of Mathematics Department in subject “Mathematical Statistics” in semester Fall 2016.

85 81 80 75 67 92 62 90 94 75 85 77 80 80 98 82 90 77 86 90 82 80 77 80
86 75 90 90 82 81 85 91 94 80 91 75 75 81 75 67 66 77 60 80 93 86 80 76
70 75 70 80 86 91 80 91 80 80 86 72

- a- How many students have marks greater than 90?
- b- How many student have marks between 65 to 80 ?
- c- Describe performance of students in the subject.

Presentation of Data

Data presentation is the method by which people summarize, organize and communicate information using a variety of tools, such as diagrams, distribution charts, histograms and graphs.

Objectives

- It condense the raw data into a form suitable for statistical analysis.
- It remove the complexities and highlights the feature of data.
- It facilitates comparison and drawing conclusion from data.
- it helps in statistical analysis by separating elements of data set into homogeneous groups and hence bring out the points of similarity and dissimilarity.

Tabular methods

- ① Frequency Distribution.

Tabular methods

- ① Frequency Distribution.
- ② Cumulative Frequency Distribution.

Tabular methods

- ① Frequency Distribution.
- ② Cumulative Frequency Distribution.
- ③ Relative Frequency Distribution.

Tabular methods

- ① Frequency Distribution.
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- ③ Relative Frequency Distribution.
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- ③ Relative Frequency Distribution.
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- ⑤ Cumulative Relative Frequency Distribution.
- ⑥ Cumulative Percentage Frequency Distribution.
- ⑦ Cross Tabulation (Contingency Table).

Frequency Distribution

A frequency distribution is a table that distributes a data set into a suitable number of categories (classes). Data presented in the form of a frequency table are called grouped data

Steps for Frequency Distribution

- 1 Decide on the number of non overlapping groups or classes. H.A Sturges proposed a rule,
$$K=1+3.22 \log N$$
 k denotes the number of classes

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- 5 Distribute the data into appropriate classes.

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- 3 Determine the class width or interval. $\text{Class Interval} = R/K$
- 4 Determine where to locate the class limit
- 5 Distribute the data into appropriate classes.
- 6 Make frequency column

Example

The following data show the marks of sixty students of Mathematics Department in subject “Mathematical Statistics” in semester Fall 2016.

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Summarize given data by Frequency Distribution

$$① K=1+3.22 \log N = 1+3.22\log(60)=6.72=7$$

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- ① $K=1+3.22 \log N = 1+3.22\log(60)=6.72=7$
- ② $R=98-60=38$
- ③ $h=R/K=38/7=5.42=5$

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- ② $R=98-60=38$
- ③ $h=R/K=38/7=5.42=5$
- ④ We start the classes from 60

Presentation of Data

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Summarize given data by Frequency Distribution

Classes	Tallies.....	Frequency
60-64		2
65-69		3
70-74		3
75-79		12
80-84		18
85-89		8
90-94		13
95-99		1

Question Construct a stem-and-leaf display for the following data.

70 72 75 64 58 83 80 82 76 75 68 65 57 78 85 72

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Question Sorting through unsolicited e-mail and spam affects the productivity of office workers. An Insight Express survey monitored office workers to determine the unproductive time per day devoted to unsolicited e-mail and spam (USA Today, November 13, 2003). The following data show a sample of time in minutes devoted to this task.

2 4 8 4 8 1 2 32 12 1 5 7 5 5 3 4 24 19 4 14

Summarize the data by constructing the following:

- A frequency distribution .
- A relative frequency distribution
- A cumulative frequency distribution
- A cumulative relative frequency distribution

Graphical methods

- 1 Histograms.

Graphical methods

- ① Histograms.
- ② .Frequency Polygon

Graphical methods

- ① Histograms.
- ② .Frequency Polygon
- ③ Frequency Curve

Graphical methods

- ① Histograms.
- ② .Frequency Polygon
- ③ Frequency Curve
- ④ Pie Diagram (Chart).

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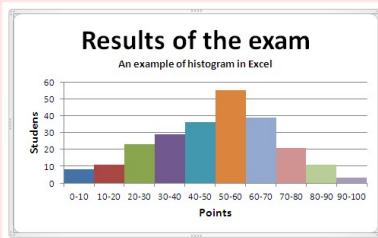
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- 5 Simple Bar Chart
- 6 Multiple Bar Chart
- 7 Component Bar Chart etc

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- 6 Multiple Bar Chart
- 7 Component Bar Chart etc
- 8 Pictogram or Ideograph

Histogram



Histogram

A histogram is a graph in which the bars are drawn adjacent to each other without any gaps. A histogram compresses a data set into a compact picture that shows the location of the mean and modes of the data and the variation in the data, especially the range. It identifies patterns in the data. This is a good aggregate graph of one variable. In order to obtain the variability in the data, it is always a good practice to start with a histogram of the data.

Steps for Histogram

- 1 Construct frequency distribution with class boundaries

Steps for Histogram

- ① Construct frequency distribution with class boundaries
- ② Measure class boundaries on horizontal axis with uniform interval.

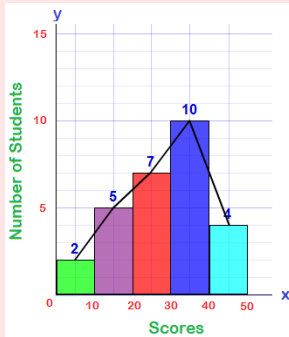
Steps for Histogram

- ① Construct frequency distribution with class boundaries
- ② Measure class boundaries on horizontal axis with uniform interval.
- ③ Measure frequency (relative frequency, percentage frequency) on vertical axis.

Steps for Histogram

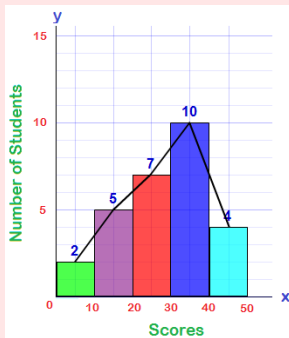
- ① Construct frequency distribution with class boundaries
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- ③ Measure frequency (relative frequency, percentage frequency) on vertical axis.
- ④ Plot the data and draw bars.

Frequency Polygon



Steps:

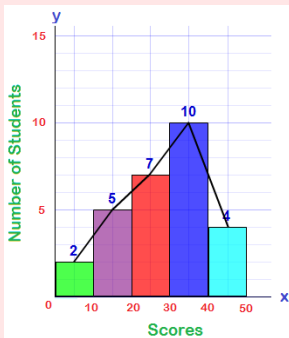
Frequency Polygon



Steps:

(1)- Draw histogram

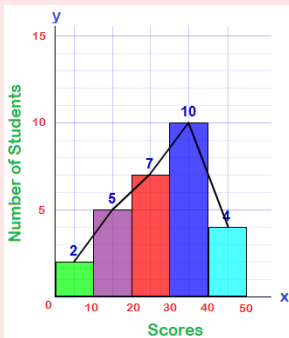
Frequency Polygon



Steps:

- (1)- Draw histogram
- (2)- Identify mid point of each bar and join these points with straight lines

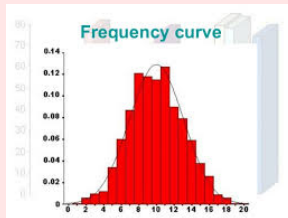
Frequency Polygon



Steps:

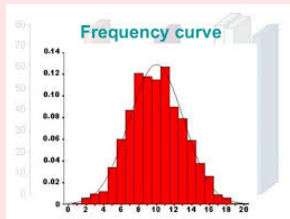
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Frequency curve



Steps:

Frequency curve



Steps:

(1)-

(2)-

Textual method

Data can be presented using paragraphs or sentences. It involves enumerating important characteristics, emphasizing significant figures and identifying the important features of data.

- **Rearrangement from lowest to highest.**
- **Stem and leaf plot.**

STEPS for Stem and Leaf

- 1 Select one or more leading digits for the stem values. The trailing digits become the leaves.
- 2 .List possible stem values in a vertical column.
- 3 Record the leaf for every observation beside the corresponding stem value.
- 4 Order the leaves from smallest to largest on each line.
- 5 Indicate the units for stems and leaves someplace in the display

Presentation of Data

Example

You are asked to present the performance of students in "Statistics". The following are the test scores :

34 42 20 50 17 9 34 43 50 18 35 43 50 23 23 35 37 38 38 39 39 38 38 39
24 29 25 26 28 27 44 44 49 48 46 45 45 46 45 46

Rearrangement of data from lowest to highest

9 17 18 20 23 23 24 25 25 26 27 28 29 34 34 3 35 37 38 38 38 38 39 39
39 42 43 43 44 44 45 45 45 46 46 46 48 49 50 50 50

Stem-and-leaf plot.

Steam	Leaves
0	9
1	7, 8
2	0, 3, 3, 4, 5, 6, 7, 8, 9
3	4, 4, 5, 5, 7, 8, 8, 8, 8, 9, 9, 9
4	2, 3, 3, 4, 4, 5, 5, 5, 6, 6, 6, 8, 9
5	0, 0, 0,

Example. Construct a stem-and-leaf display for the following data.

70 72 75 64 58 83 80 82 76 75 68 65 57 78 85 72

Example. Construct a stem-and-leaf display for the following data.

11.3 9.6 10.4 7.5 8.3 10.5 10.0 9.3 8.1 7.7 7.5 8.4 6.3 8.8

Presentation of Data

The Brunswick Research Organization surveyed 50 randomly selected individuals and asked them the primary way they received the daily news. Their choices were via newspaper (N), television (T), radio (R), or Internet (I).

N N T T T I R R I T I N R R I N N I T N I R T T T T N R R I R R I N T
R T I I T T I N T T I R N R T

Construct a categorical frequency distribution , Pie chart and bar chart, for the data.

Presentation of data

Consider the following data.

16 20 22 21 18 19 22 25 16 17 24 24 25 19 16 19 18 17 21 13 16 17 18 23
24 20 23 16 20 20 24 26 15 22 24 20 22 24 22 20 15 14 16 17 18

- Develop a frequency distribution
- Make Histogram and frequency Polygon.

Example

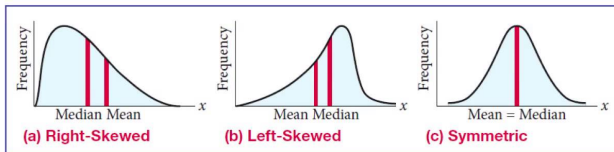
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Summarize data by frequency distribution.

Skewed and Symmetric Distributions

- Symmetric Data
 - Data sets whose values are evenly spread around the center.
- Skewed Data
 - Data sets that are not symmetric



Demerits of Presentation of Data

Although presentation of data in different forms (textual, Tabular or Graphical) make raw data more meaningful. But these methods fail to identify four major properties that describe any set of quantitative data. These four major properties are:

- 1 **Central Tendency:** The numerical values of observation (also called central value) around which most numerical values of data show a tendency to cluster or group.

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- 3 **Skewness:** The extent of departure of values from symmetrical (Bell Shape) Distribution.

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- 3 **Skewness:** The extent of departure of values from symmetrical (Bell Shape) Distribution.
- 4 **Kurtosis.** The extent of peakness of symmetrical distribution.

Descriptive Measure or Summary Measure

The four properties are used to extract and summarize major feature of the data set by the application of certain statistical methods called descriptive measures or summary measures: These Measures are:

- ① **Measures of Central Tendency:**
- ② **Measures of Variation or Dispersion:**
- ③ **Measures of Skewness:**
- ④ **Measures of Kurtosis.**

Measures of Central Tendency

We can easily present things as we wish them, to be..." (Aesop)

If at first you do not succeed, you are just about average. (Bill Cosby)

Objectives

- 1 Summarize the data

Measures of Central Tendency

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- 2 Comparison between two or more data sets

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Objectives

- 1 Summarize the data
- 2 Comparison between two or more data sets
- 3 Provide base for calculating other statistical measures

Measures of Central Tendency

Measure of Central Tendency

- 1 Arithmetic Mean

Measures of Central Tendency

Measure of Central Tendency

- ① Arithmetic Mean
- ② Geometric Mean

Measures of Central Tendency

Measure of Central Tendency

- ① Arithmetic Mean
- ② Geometric Mean
- ③ Harmonic Mean

Measures of Central Tendency

Measure of Central Tendency

- ① Arithmetic Mean
- ② Geometric Mean
- ③ Harmonic Mean
- ④ Median

Measures of Central Tendency

Measure of Central Tendency

- ① Arithmetic Mean
- ② Geometric Mean
- ③ Harmonic Mean
- ④ Median
- ⑤ Mode

Criteria or Properties of Good Measures of Central Tendency

- 1 Easily calculated and understandable.

Measures of Central Tendency

Criteria or Properties of Good Measures of Central Tendency

- ① Easily calculated and understandable.
- ② Based on all values

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- ② Based on all values
- ③ Does not effected by extreme values

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- ④ Rigidly define
- ⑤ It can be used for further statistical analysis
- ⑥ Sampling Stability

Measures of Central Tendency

Arithmetic Mean

Arithmetic mean is a value which is obtained by dividing the sum of all values by their number.

Let $x_1, x_2, \dots, x_N$ be the value of population of size N . The population mean denoted by μ defined as

$$\mu = \frac{x_1 + x_2 + \dots + x_N}{N}$$

Let $x_1, x_2, \dots, x_n$ be the value of sample of size n . The population mean denoted by $\bar{x}$ defined as

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks.

54, 60, 70, 90, 60, 60,

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks.

54, 60, 70, 90, 60, 60,

Find the arithmetic mean.

Solution

$$\bar{x} = \frac{54 + 60 + 70 + 90 + 60 + 60}{6} = \frac{394}{6} = 65.67$$

Measures of Central Tendency

Geometric Mean

Geometric mean is a value which is obtained by N^{th} under-root of product of all N values

Or

Geometric mean is the antilog of arithmetic of log of the values.

Let $x_1, x_2, \dots, x_N$ be the value of population of size N . The Geometric mean denoted by $G.M$ defined as

$$G.M (x_1.x_2...x_N)^{\frac{1}{N}}$$

or

$$G.M = Antilog \left(\frac{\log x_1 + \log x_2 + \dots + \log x_N}{N} \right)$$

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks.

54, 60, 70, 90, 60, 60,

Find the Geometric mean.

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks.

54, 60, 70, 90, 60, 60,

Find the Geometric mean.

Solution

$$G.M = (54.60.70.90.60.60)^{\frac{1}{6}} = \dots$$

Or

$$G.M = \frac{\log 54 + \log 60 + \log 70 + \log 90 + \log 60 + \log 60}{6} = \dots$$

Measures of Central Tendency

Example

The return on investment earned by Atkins Construction Company for four successive years was: 30 percent, 20 percent, -40 percent, and 200 percent. What is the geometric mean rate of return on investment?

Harmonic Mean

Harmonic mean is the reciprocal of arithmetic mean of reciprocal of values. Let $x_1, x_2, \dots, x_N$ be the value of population of size N . The Harmonic mean denoted by $H.M$ defined as

$$H.M = \frac{N}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_N}}$$

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks.

54, 60, 70, 90, 60, 60,

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks.

54, 60, 70, 90, 60, 60,

Find the Harmonic mean.

Solution

$$H.M = \frac{6}{\frac{1}{54} + \frac{1}{60} + \frac{1}{70} + \frac{1}{90} + \frac{1}{60} + \frac{1}{60}} = \dots$$

Measures of Central Tendency

Median

Median is a value which divide an array data into two equal parts. Median is a position measure and equal number of values lie above and below the median value. Let $x_1, x_2, \dots, x_N$ be the value of population of size N . Median is denoted by $\hat{X}$

$$\hat{X} = \left(\frac{N+1}{2} \right) \text{th value, if } N \text{ is odd}$$

$$\hat{X} = \frac{1}{2} \left[\left(\frac{N}{2} \right) \text{th value} + \left(\frac{N}{2} + 1 \right) \text{th value} \right], \text{ if } N \text{ is even}$$

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks. 54, 60, 70, 90, 65, 60, Find the Median Marks.

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks. 54, 60, 70, 90, 65, 60, Find the Median Marks.

Solution

Array data: 54, 60 , 60, 65, 70, 90

$$\hat{X} = \frac{1}{2} \left[\left(\frac{6}{2} \right) \text{th value} + \left(\frac{6}{2} + 1 \right) \text{th value} \right]$$

$$\hat{X} = \frac{1}{2} [3\text{rd value} + 4\text{th value}] = \frac{1}{2} [60 + 65] = 62.5$$

Measures of Central Tendency

Mode

Mode is the value(s) which occur most frequent number of time in data set It is denoted by $\hat{X}$. If any value does not repeat more then once. Then we say, Mode does not exit. If more then one values repeat equal number of times. Then these values will be mode

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks. 54, 60, 70, 90, 65, 60, Find the Mode.

Measures of Central Tendency

Example

The following data represents the marks of 6 students selected at random from class of 40 students in the subject “ Statistics and probability” out of 100 marks. 54, 60, 70, 90, 65, 60, Find the Mode.

Solution

$$\text{Mode} = \hat{X} = 60$$

Measures of Central Tendency

Example

Erika operates a website devoted to providing information and support for persons who are interested in organic gardening. According to the hit counter that records daily visitors to her site, the numbers of visits during the past 8 days have been as follows:

70, 37, 55, 63, 59, 68, 56, 54.

Determine the mean, Geometric mean, Harmonic mean, Mode and Median

Measures of Central Tendency

Example

A social scientist for a children's advocacy organization has randomly selected 10 Saturday-morning television cartoon shows and carried out a content analysis in which he counts the number of incidents of verbal or physical violence in each. For the 10 cartoons examined, the counts were as follows:

27, 12, 16, 22, 15, 30, 14, 30, 11, 21.

Find the mean, Geometric mean, Harmonic mean, Mode and Median

Measures of Central Tendency

Example

A social scientist for a children's advocacy organization has randomly selected 10 Saturday-morning television cartoon shows and carried out a content analysis in which he counts the number of incidents of verbal or physical violence in each. For the 10 cartoons examined, the counts were as follows:

27, 12, 16, 22, 15, 30, 14, 30, 11, 21.

Find the mean, Geometric mean, Harmonic mean, Mode and Median

Measures of Central Tendency

Example

The percent increase in sales for the last 4 years at Combs Cosmetics were: 4.91, 5.75, 8.12, and 21.60. (a) Find the geometric mean percent increase. (b) Find the arithmetic mean percent increase. (c) Is the arithmetic mean equal to or greater than the geometric mean?

Measures of Central Tendency

Assignment/Home Work

Find the arithmetic mean, Geometric mean, Harmonic mean, Mode and Median using data given in question number 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.10, 1.11, 1.12

Measures of Central Tendency

Assignment

Problem 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7 and some other problem on these topics

Measures of Dispersion

Objectives

- 1 Summarize the data

Measures of Dispersion

Objectives

- 1 Summarize the data
- 2 Comparison between two or more data sets

Measures of Dispersion

Objectives

- 1 Summarize the data
- 2 Comparison between two or more data sets
- 3 Provide base for calculating other statistical measures

Measures of Dispersion

Measures of Dispersion

① Absolute Measures of Dispersion

Measures of Dispersion

Measures of Dispersion

- ① Absolute Measures of Dispersion
- ② Relative Measures of Dispersion

Measures of Dispersion

Measures of Dispersion

- ① Absolute Measures of Dispersion
- ② Relative Measures of Dispersion

Criteria or Properties of Good Measure of Dispersion

- 1 Easily calculated and understandable.

Measures of Dispersion

Criteria or Properties of Good Measure of Dispersion

- ① Easily calculated and understandable.
- ② Based on all values

Measures of Dispersion

Criteria or Properties of Good Measure of Dispersion

- ① Easily calculated and understandable.
- ② Based on all values
- ③ Does not effected by extreme values

Criteria or Properties of Good Measure of Dispersion

- ① Easily calculated and understandable.
- ② Based on all values
- ③ Does not effected by extreme values
- ④ Rigidly define

Criteria or Properties of Good Measure of Dispersion

- ① Easily calculated and understandable.
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- ⑤ It can be used for further statistical analysis

Criteria or Properties of Good Measure of Dispersion

- ① Easily calculated and understandable.
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- ④ Rigidly define
- ⑤ It can be used for further statistical analysis
- ⑥ Sampling Stability

Measures of Dispersion

Absolute Measures of Dispersion

① Range

Measures of Dispersion

Absolute Measures of Dispersion

- 1 Range
- 2 Mean Deviation

Measures of Dispersion

Absolute Measures of Dispersion

- ① Range
- ② Mean Deviation
- ③ Variance

Measures of Dispersion

Absolute Measures of Dispersion

- ① Range
- ② Mean Deviation
- ③ Variance
- ④ Standard Deviation

Measures of Dispersion

Relative Measures of Dispersion

- ① Coefficient of Range

Measures of Dispersion

Relative Measures of Dispersion

- ① Coefficient of Range
- ② Coefficient Mean Deviation

Measures of Dispersion

Relative Measures of Dispersion

- ① Coefficient of Range
- ② Coefficient Mean Deviation
- ③ Coefficient Variation



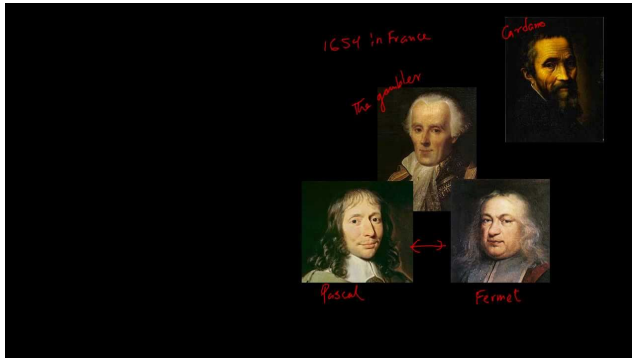




Probability



Probability



The Pascal – Fermat correspondence of 1654



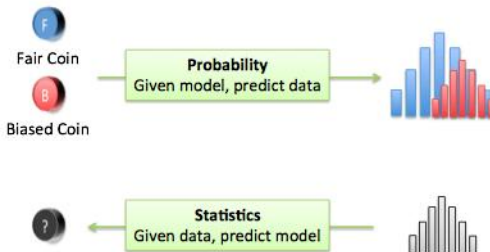
- Often cited in histories of mathematics as the origin of probability theory.



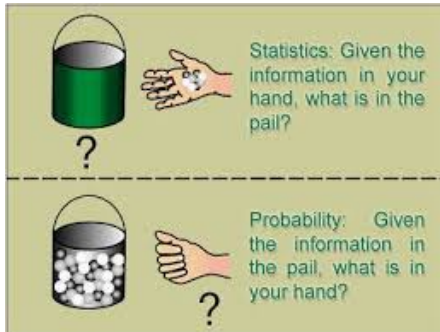
"It is remarkable that a science which began with the consideration of games of chance should have become the most important object of human knowledge."

Pierre Simon de Laplace

Probability & Statistics



Probability



THE MEANING OF PROBABILITY

Probability is used to describe **RANDOM** or **CHANCES** of events to occur.

Every day we are faced with probability statements involving the words:



1. What is the **likelihood** that X will occur?
2. What is the **chance** that Brazil will win the 2014 World Cup?

Probability theory provides a mathematical model for the study of randomness and uncertainty

Experiment



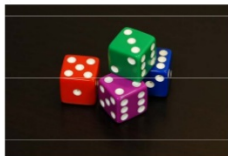
Experiment



Experiment

RANDOM EXPERIMENTS

- A random experiment is an experiment in which the outcome varies in an unpredictable fashion when the experiment is repeated under the same conditions.



Probability and Random Variables, Lecture 2, by Dr. Sobia Baig

4

Sample Space

Set of All possible outcomes of random experiment is known as sample space.



Table 4.1 Examples of Experiments, Outcomes, and Sample Spaces

Experiment	Outcomes	Sample Space
Toss a coin once	Head, Tail	$S = \{\text{Head, Tail}\}$
Roll a die once	1, 2, 3, 4, 5, 6	$S = \{1, 2, 3, 4, 5, 6\}$
Toss a coin twice	HH, HT, TH, TT	$S = \{HH, HT, TH, TT\}$
Play lottery	Win, Lose	$S = \{\text{Win, Lose}\}$
Take a test	Pass, Fail	$S = \{\text{Pass, Fail}\}$
Select a student	Male, Female	$S = \{\text{Male, Female}\}$

Sample Space

Set of All possible outcomes of random experiment is known as sample space.

Table A

Outcome of first die	Outcome of second die					
	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

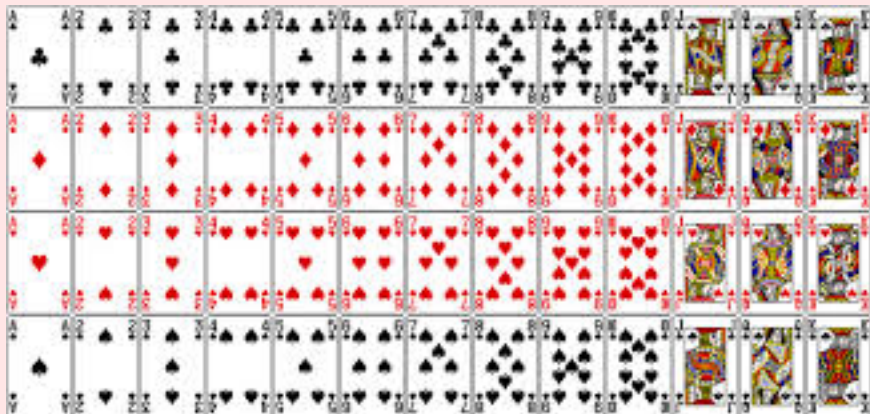
Sample Space

Set of All possible outcomes of random experiment is known as sample space.

Probability

Cards

Red card include Heart and Diamond. Black Card Include Club and Spade



Event and its Types

Subset of sample space is called event.

Types of event

- 1 Null event

Event and its Types

Subset of sample space is called event.

Types of event

- 1 Null event
- 2 Sure Event

Event and its Types

Subset of sample space is called event.

Types of event

- 1 Null event
- 2 Sure Event
- 3 Simple Event

Event and its Types

Subset of sample space is called event.

Types of event

- 1 Null event
- 2 Sure Event
- 3 Simple Event
- 4 Compound Event

Event and its Types

Subset of sample space is called event.

Types of event

- 1 Null event
- 2 Sure Event
- 3 Simple Event
- 4 Compound Event
- 5 Mutually Exclusive Event

Event and its Types

Subset of sample space is called event.

Types of event

- 1 Null event
- 2 Sure Event
- 3 Simple Event
- 4 Compound Event
- 5 Mutually Exclusive Event
- 6 Exhaustive Event

Event and its Types

Subset of sample space is called event.

Types of event

- 1 Null event
- 2 Sure Event
- 3 Simple Event
- 4 Compound Event
- 5 Mutually Exclusive Event
- 6 Exhaustive Event
- 7 Equally Likely Events

Example

Some people are in favor of reducing federal taxes to increase consumer spending and others are against it. Two persons are selected and their opinions are recorded. List the possible outcomes.

Example

Some people are in favor of reducing federal taxes to increase consumer spending and others are against it. Two persons are selected and their opinions are recorded. List the possible outcomes.

Example

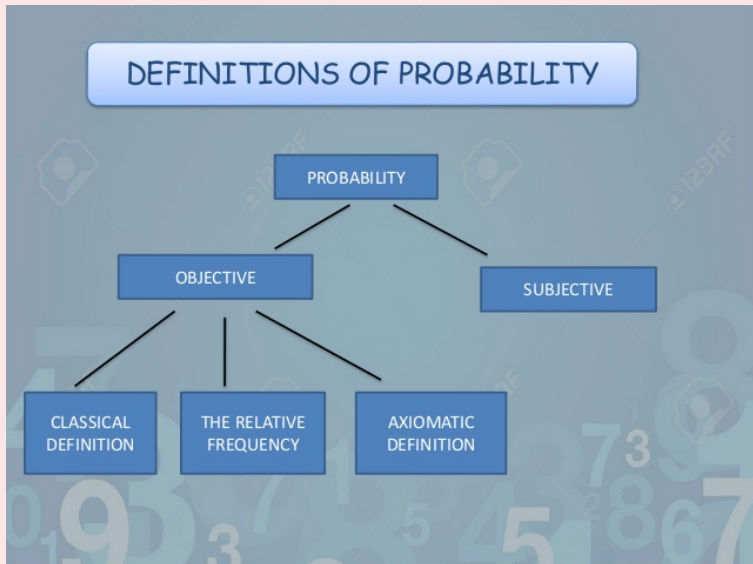
A quality control inspector selects a part to be tested. The part is then declared acceptable, repairable, or scrapped. Then another part is tested. List the possible outcomes of this experiment regarding two parts.

Example

Example

Example

Approaches of Probability



Approaches of Probability

Subjective Approach

Subjective probability is a probability derived from an individual's personal judgment about whether a specific outcome is likely to occur. It contains no formal calculations and only reflects the subject's opinions and past experience. Subjective probabilities differ from person to person, and they contain a high degree of personal bias.

Objective Approach

The probability that an event will occur based on an analysis in which each measure is based on a recorded observation is objective probability approach

Classical or Prior definition

A random experiment can produce n equally likely and mutually exclusive possible outcomes and m outcomes are favorable to event A . Then probability of event A is defined as:

(CLASSICAL DEFINITION OF PROBABILITY)

$$P(\text{Event}) = \frac{\# \text{ Favorable Outcomes}}{\# \text{ Possible Outcomes}}$$

Posterior or relative frequency definition

Axiomatic Definition

Axioms :

- ▣ $P(A) \geq 0$ (Probability cannot be negative)
- ▣ $P(S) = 1$ (Probability of everything is 1)
- ▣ If the events A & B are mutually exclusive, then
 - $P(A \text{ and } B) = P(A \cap B) = \emptyset$
 - Probability of occurrence of both events is null
 - $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$
 - Probability of occurrence of either of the events

Probability of Event

Example

Two fair coins are tossed. What is the probability of getting one heads and one tails?

Probability of Event

Example

Two fair coins are tossed. What is the probability of getting one heads and one tails?

Example

A coin is tossed twice. What is the probability that at least one head occurs

Probability of Event

Example

Two fair coins are tossed. What is the probability of getting one heads and one tails?

Example

A coin is tossed twice. What is the probability that at least one head occurs

Example

In a poker hand consisting of 5 cards, find the probability of holding 2 aces and 3 jacks

Probability of Event

Example

Two fair coins are tossed. What is the probability of getting one heads and one tails?

Example

A coin is tossed twice. What is the probability that at least one head occurs

Example

In a poker hand consisting of 5 cards, find the probability of holding 2 aces and 3 jacks

Example

A bag contains 6 red balls, 5 yellow balls and 3 green balls. A ball is drawn at random. What is the probability that the ball is: (a) green, (b) not yellow, (c) red or yellow?

Example

A fair coin is tossed three times. What is the probability that the sequence will be heads, tails, heads?

Example

A boxcar contains six complex electronic systems. Two of the six are to be randomly selected for thorough testing and then classified as defective or not defective.

- a- If two of the six systems are actually defective, find the probability that at least one of the two systems tested will be defective. Find the probability that both are defective.
- b- If four of the six systems are actually defective, find the probabilities indicated in part (a)

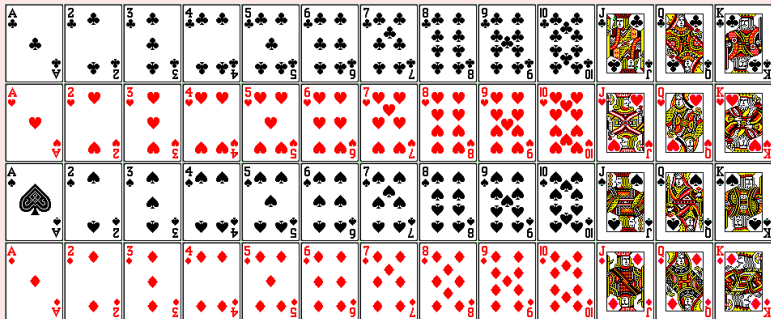
Example

A committee of 3 persons is to be selected randomly from a group of 5 men and 3 women.

- (a) Find the probability that the committee consists of 1 men and women.
- (b) Find the probability that the committee consists of all women

Probability

Card



Counting Rule for sample points

Counting sample points

- Rule of Multiplication
- Rule of Permutation
- Rule of Combination

Counting Rule for Sample Points

Multiplication Rule

If an operation can be performed in n_1 ways, and if for each of these a second operation can be performed in n_2 ways, and for each of the first two a third operation can be performed in n_3 ways, and so forth, then the sequence of k operations can be performed in $n_1 n_2 \dots n_k$ ways.

Counting Rule for Sample Points

Permutation

A permutation is an arrangement of all or part of a set of objects.

The number of permutation of n distinct objects taken r at a time is

Counting Rule for Sample Points

Combination

The number of combinations of n distinct objects taken r at a time is

Question 11

A boxcar contains six complex electronic systems. Two of the six are to be randomly selected for thorough testing and then classified as defective or not defective.

- a- If two of the six systems are actually defective, find the probability that at least one of the two systems tested will be defective. Find the probability that both are defective.
- b- If four of the six systems are actually defective, find the probabilities indicated in part (a)

Question 12

A retailer sells only two styles of stereo consoles, and experience shows that these are in equal demand. Four customers in succession come into the store to order stereos. The retailer is interested in their preferences.

- a- List the possibilities for preference arrangements among the four customers (that is, list the sample space).
- b- Assign probabilities to the sample points.
- c- Let A denote the event that all four customers prefer the same style. Find $P(A)$

Question 12

A retailer sells only two styles of stereo consoles, and experience shows that these are in equal demand. Four customers in succession come into the store to order stereos. The retailer is interested in their preferences.

- a- List the possibilities for preference arrangements among the four customers (that is, list the sample space).
- b- Assign probabilities to the sample points.
- c- Let A denote the event that all four customers prefer the same style. Find $P(A)$

Question 13

A committee of 3 persons is to be selected randomly from a group of 5 men and 3 women.

- (a) Find the probability that the committee consists of 1 men and women.
- (b) Find the probability that the committee consists of all women

Question 14

In a tank containing 10 fishes, there are three yellow and seven black fishes. We select three fishes at random.

- (a) What is the probability that exactly one yellow fish gets selected?
- (b) What is the probability that at most one yellow fish gets selected?
- (c) What is the probability that at least one yellow fish gets selected?

Question 14

In a tank containing 10 fishes, there are three yellow and seven black fishes. We select three fishes at random.

- (a) What is the probability that exactly one yellow fish gets selected?
- (b) What is the probability that at most one yellow fish gets selected?
- (c) What is the probability that at least one yellow fish gets selected?

Question 15

A package of 15 apples contains two defective apples. Four apples are selected at random.

- (a) Find the probability that none of the selected apples is defective.
- (b) Find the probability that at least one of the selected apples is defective

Question 16

A homeowner wants to repaint her home and install new carpets (no store where she live sells both paint and carpet). She plans to get the services from the stores where she buys the paint and carpet. Suppose there are 12 paint stores with painting service available and 15 carpet stores with installation services available in that city. In how many ways can she choose these two stores

Question 16

A homeowner wants to repaint her home and install new carpets (no store where she live sells both paint and carpet). She plans to get the services from the stores where she buys the paint and carpet. Suppose there are 12 paint stores with painting service available and 15 carpet stores with installation services available in that city. In how many ways can she choose these two stores

Question 17

From an urn containing 15 white, 7 black, and 8 yellow balls a sample of 3 balls is drawn at random. Find the probability that

- (a) All three balls are yellow.
- (b) All three balls are of the same color.
- (c) All three balls are of different colors.

Question 18

A box of manufactured items contains 12 items, of which four are defective. If three items are drawn at random without replacement, what is the probability that

- (a) The first one is defective and the rest are good?
- (b) Exactly one of the three is defective?

Addition law of probability

If A and B are event defined in a sample space S. Probability that event A or event B (at least on of them) will occur is denoted by $p(A \cup B)$ defined as

$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

Proof

Question 19

A single 6-sided die is rolled. What is the probability of rolling a 2 or a 5

Question 19

A single 6-sided die is rolled. What is the probability of rolling a 2 or a 5

Question 20

A single 6-sided die is rolled. What is the probability of rolling a 2 or a 5

Question 21

A spinner has 4 equal sectors colored yellow, blue, green, and red. What is the probability of landing on red or blue after spinning this spinner?

Question 21

A spinner has 4 equal sectors colored yellow, blue, green, and red. What is the probability of landing on red or blue after spinning this spinner?

Question 22

In a math class of 30 students, 17 are boys and 13 are girls. On a unit test, 4 boys and 5 girls made an A grade. If a student is chosen at random from the class, what is the probability of choosing a girl or an A student?

Question 23

A bag contains 18 coloured marbles: 4 are coloured red, 8 are coloured yellow and 6 are coloured green. A marble is selected at random. What is the probability that the ball chosen is either red or green?

Question 23

A bag contains 18 coloured marbles: 4 are coloured red, 8 are coloured yellow and 6 are coloured green. A marble is selected at random. What is the probability that the ball chosen is either red or green?

Question 24

A lot consists of 10 good articles, 4 with minor defects, and 2 with major defects. One article is chosen at random. Find the probability that:

- (a) it has no defects
- (b) it has no major defects
- (c) it is either good or has major defects

Question 25

A bag contains 20 balls, 3 are coloured red, 6 are coloured green, 4 are coloured blue, 2 are coloured white and 5 are coloured yellow. One ball is selected at random. Find the probabilities of the following events.

- (a) A red marble
- (b) A green marble
- (c) A white marble
- (d) red or white marble
- (e) green or white marble
- (f) green, red or white marble

Question 25

A bag contains 20 balls, 3 are coloured red, 6 are coloured green, 4 are coloured blue, 2 are coloured white and 5 are coloured yellow. One ball is selected at random. Find the probabilities of the following events.

- (a) A red marble
- (b) A green marble
- (c) A white marble
- (d) red or white marble
- (e) green or white marble
- (f) green, red or white marble

Question 26

In a group of 101 students 40 are juniors, 50 are female, and 22 are female juniors. Find the probability that a student picked from this group at random is either a junior or female.

Question 27

A single card is chosen at random from a standard deck of 52 playing cards. What is the probability of choosing a king or a club?

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A single card is chosen at random from a standard deck of 52 playing cards. What is the probability of choosing a king or a club?

Question-28

Hydraulic landing assemblies coming from an aircraft rework facility are each inspected for defects. Historical records indicate that 8% have defects in shafts only, 6% have defects in bushings only, and 2% have defects in both shafts and bushings. One of the hydraulic assemblies is selected randomly. What is the probability that the assembly has

- (a) a bushing defect?
- (b) a shaft or bushing defect?
- (c) exactly one of the two types of defects?
- (d) neither type of defect?

Question-29

If $P(A) = 0.24$, $P(B) = 0.67$, and $P(AB) = 0.09$, find

- (a) $P(A \cup B)$
- (b) $P((A \cup B)^c)$
- (c) $P(A^c \cup B^c)$

Question-29

If $P(A) = 0.24$, $P(B) = 0.67$, and $P(AB) = 0.09$, find

- (a) $P(A \cup B)$
- (b) $P((A \cup B)^c)$
- (c) $P(A^c \cup B^c)$

Question-30

A sample of four electronic components is taken from the output of a production line. The probabilities of the various outcomes are calculated to be: $\Pr [0 \text{ defectives}] = 0.6561$, $\Pr [1 \text{ defective}] = 0.2916$, $\Pr [2 \text{ defectives}] = 0.0486$, $\Pr [3 \text{ defectives}] = 0.0036$, $\Pr [4 \text{ defectives}] = 0.0001$. What is the probability of at least one defective?

Question-31

If one card is drawn from a well-shuffled bridge deck of 52 playing cards (13 of each suit), what is the probability that the card is a queen or a heart?

Question-31

If one card is drawn from a well-shuffled bridge deck of 52 playing cards (13 of each suit), what is the probability that the card is a queen or a heart?

Question-32

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Question-33 Assign:

The class registrations of 120 students are analyzed. It is found that: 30 of the students do not take any of Applied Mechanics, Chemistry, or Computers. 15 of them take only Applied Mechanics. 25 of them take Chemistry and Computers but not Applied Mechanics. 20 of them take Applied Mechanics and Computers but not Chemistry. 10 of them take all three of Applied Mechanics, Chemistry, and Computers. A total of 45 of them take Chemistry. 5 of them take only Chemistry.

- How many of the students take Applied Mechanics and Chemistry but not Computers?
- How many of the students take only Computers?
- What is the total number of students taking Computers?
- If a student is chosen at random from those who take neither Chemistry nor Computers, what is the probability that he or she does not take Applied Mechanics either?
- If one of the students who take at least two of the three courses is chosen at random, what is the probability that he or she takes all three

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Question-34

Past records show that 4 of 135 parts are defective in length, 3 of 141 are defective in width, and 2 of 347 are defective in both. Use these figures to estimate probabilities of the individual events assuming that defects occur independently in length and width. a) What is the probability that a part produced under the same conditions will be defective in length or width or both?

b) What is the probability that a part will have neither defect?

c) What are the fair odds against a defect (in length or width or both)?

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Question-35

In a group of 72 students, 14 take neither English nor chemistry, 42 take English and 38 take chemistry. What is the probability that a student chosen at random from this group takes:

a) both English and chemistry?

b) chemistry but not English?

Question-36

A fair six-sided die is tossed twice. What is the probability that a five will occur at least once

Conditional Probability

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

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Question-37

A math teacher gave her class two tests. 25% of the class passed both tests and 42% of the class passed the first test. What percent of those who passed the first test also passed the second test

Question-38

A jar contains black and white marbles. Two marbles are chosen without replacement. The probability of selecting a black marble and then a white marble is 0.34, and the probability of selecting a black marble on the first draw is 0.47. What is the probability of selecting a white marble on the second draw, given that the first marble drawn was black?

Question-39 Assignment

At Kennedy Middle School, the probability that a student takes Technology and Spanish is 0.087. The probability that a student takes Technology is 0.68. What is the probability that a student takes Spanish given that the student is taking Technology?

A laboratory blood test is 98% effective in detecting a certain disease if the person has the disease (sensitivity). However, the test also yields a “false positive” result for 0.5% of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.005, the test result will show positive.) Assume that 2% of the population actually has this disease (prevalence). What is the probability a person has the disease given that the test result is positive?

Dependent and Independent Event

Two events are said to be independent if the result of the second event is not affected by the result of the first event. Otherwise, events are called dependent events.

Question-40

A card is elected at random from an ordinary deck of 52 playing cards. If E is the event that the selected card is an ace and F is the event that it is a spade, show that E and F are independent events

Multiplication Rule

If A and B be the two events defined in a sample space S. The probability that event A and event B or both of them will occur is denoted by $P(AB)$ or

$$P(AB) = P(A)P(B|A)$$

$$P(ABC) = P(A)P(B|A)P(C|AB)$$

$$P(ABCD) = P(A)P(B|A)P(C|AB)P(D|ABC)$$

Question

Four individuals have responded to a request by a blood bank for blood donations. None of them has donated before, so their blood types are unknown. Suppose only type O+ is desired and only one of the four actually has this type. If the potential donors are selected in random order for typing, what is the probability that at least three individuals must be typed to obtain the desired type?

Question for Odd Rows

It is known that 30% of a certain company's washing machines require service while under warranty, whereas only 10% of its dryers need such service. If someone purchases both a washer and a dryer made by this company, what is the probability that both machines need warranty service?

Question For Even Rows

An aircraft seam requires 25 rivets. The seam will have to be reworked if any of these rivets is defective. Suppose rivets are defective independently of one another, each with the same probability. If 20% of all seams need reworking, what is the probability that a rivet is defective?

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